

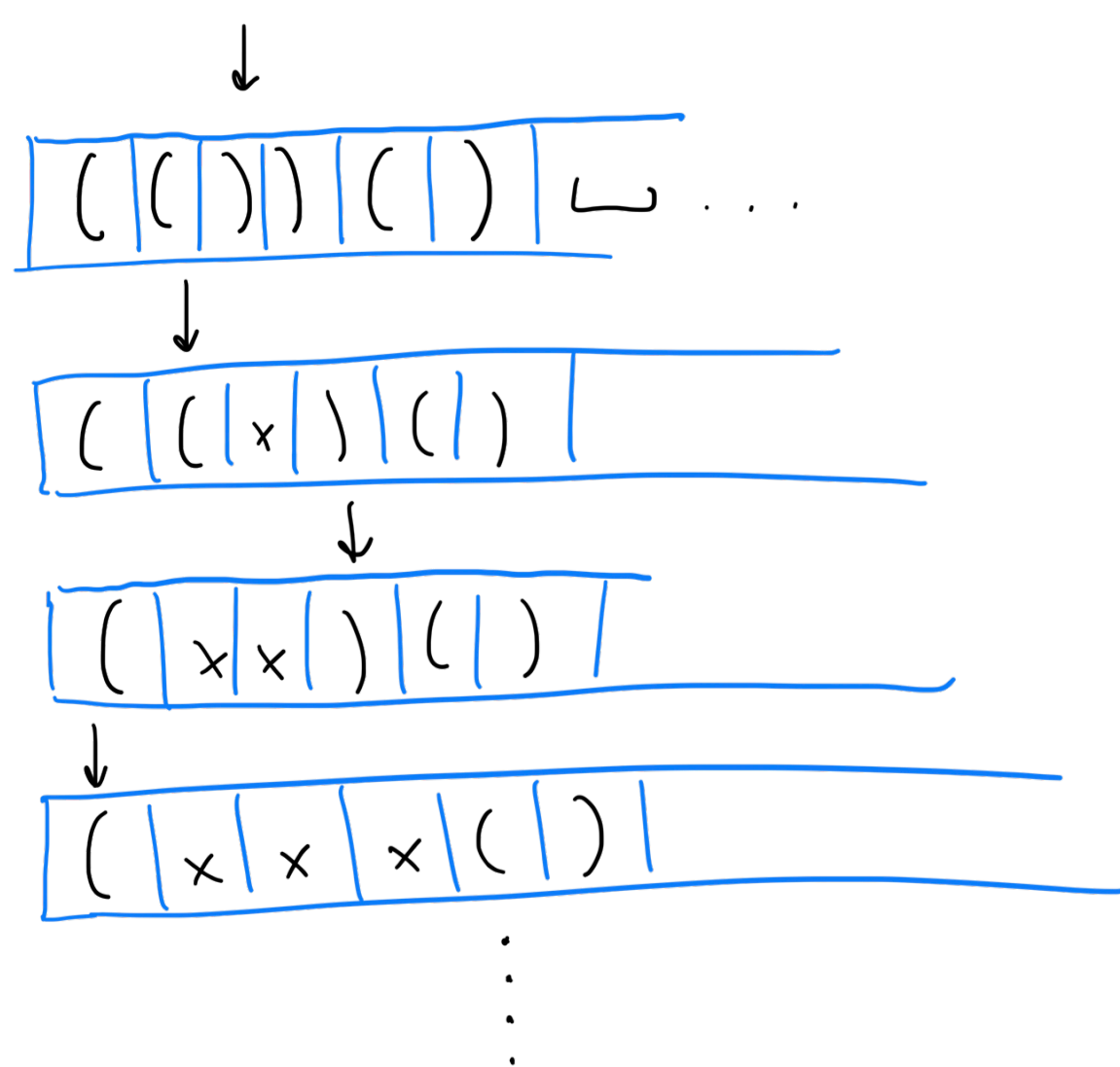
Extra Problems

- Create a TM that accepts balanced parentheses.

ie. $()$, $()()$, $(())$, $(())()$, etc.

Give an example via a snapshot of configurations (snapshot history).

1. For each $)$, find the closest unmatched $($. Mark both as matched.
2. Repeat step 2 until no more unmatched $)$'s exist.
3. Check the tape for any unmatched $($'s. If found, **reject**. Otherwise, **accept**.



Question 4.16. Let $A = \{ \langle R \rangle \mid R \text{ is a regex that accepts some string with substring } III \}$. Show A is decidable.

$T =$ "On input $\langle R \rangle$,
Test all strings containing III
to see if any is accepted by R .
If so, **accept**.

~~If all strings are exhausted, **reject**. "□"~~
Not possible!

T is a mere recognizer.

M is a decider...

$M =$ "On input $\langle R \rangle$:

Let $C =$ be an NFA.

Note that $L(C) =$ all strings with III .

Let $S = L(R) \cap L(C)$.

Convert S to a DFA D s.t. $L(D) = S$.

By Theorem 4.4, there exists a

decider E_{DFA} . If E_{DFA} decides that $L(D) = \emptyset$,
reject. Otherwise, **accept**. "□"